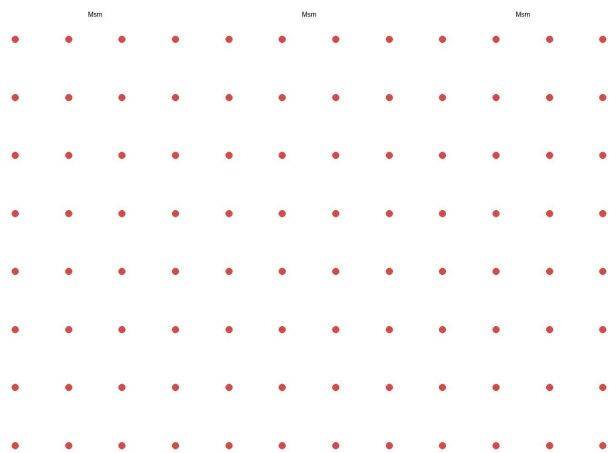
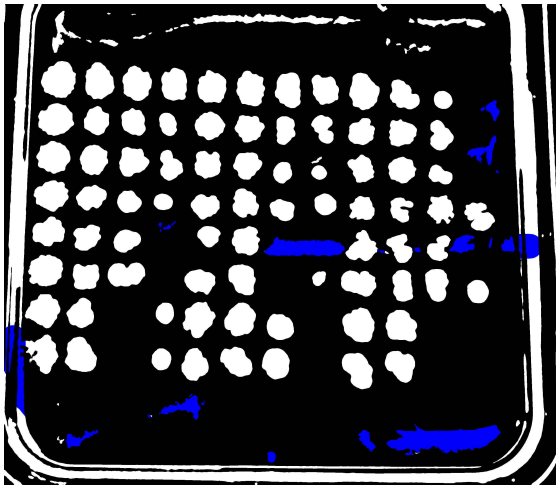


Ecoli-1_NoDrug_IMG_3310.JPG Log

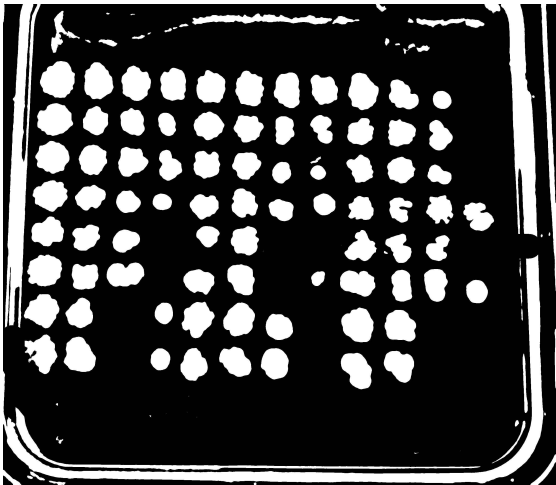


Filename: Ecoli-1_NoDrug_IMG_3310.JPG
Additive: None
X Dilution: 10
Y Dilution: 2
Strains: Msm, Msm, Msm
Column Indexes: [[0, 1, 2, 3], [4, 5, 6, 7], [8, 9, 10, 11]]
Gap Between Strains: False
Threshold: 2
Minimum Area: 15
Block Size: 179

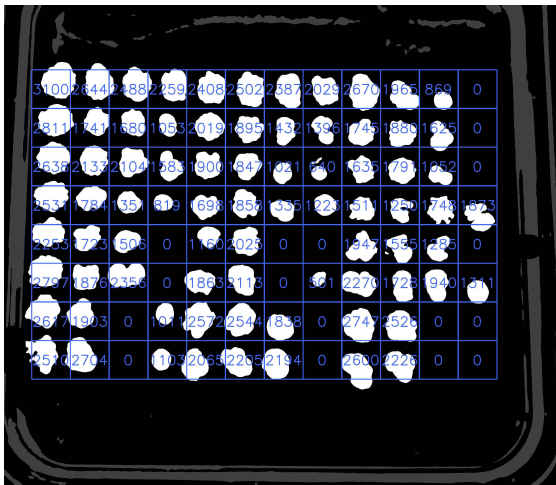
Preview Image



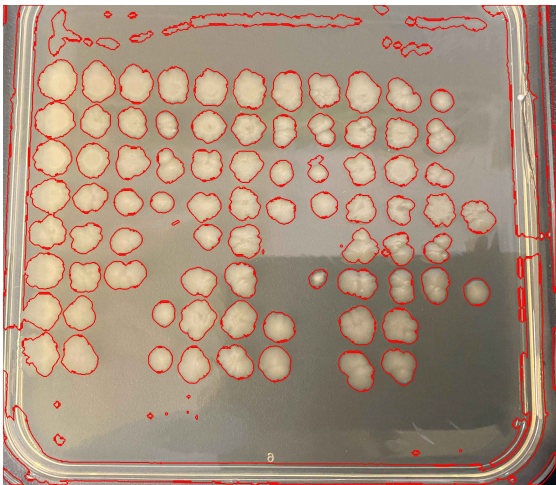
Tool Usage: Red(+) Blue(-)



Binary Image



Grid Image



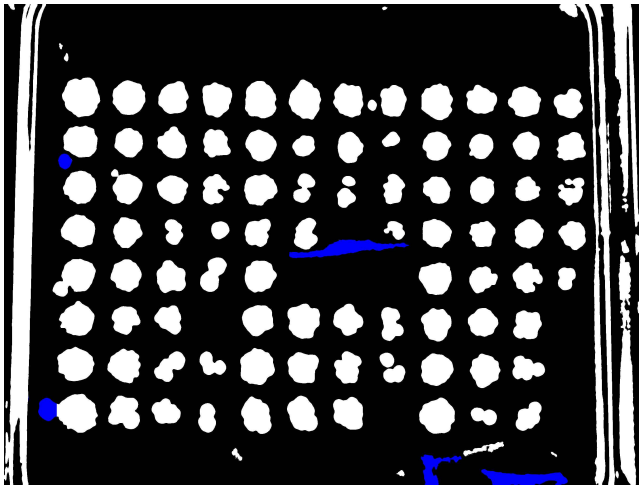
Contours Image

Ecoli-2_Kanamycin_IMG_3311.JPG Log

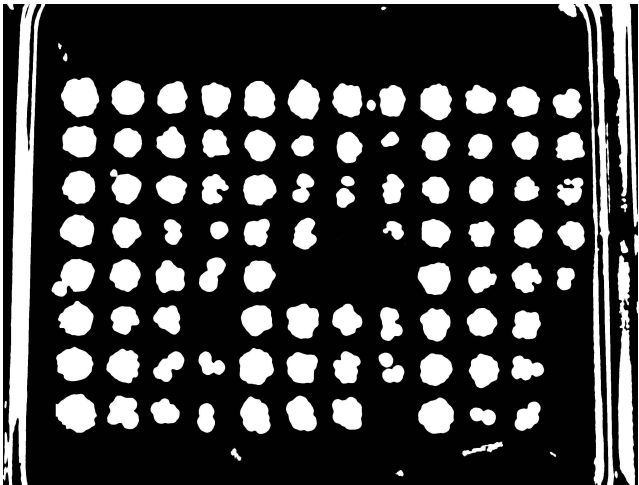


Filename: Ecoli-2_Kanamycin_IMG_3311.JPG
Additive: Kanamycin
X Dilution: 10
Y Dilution: 2
Strains: Msm, Msm, Msm
Column Indexes: [[0, 1, 2, 3], [4, 5, 6, 7], [8, 9, 10, 11]]
Gap Between Strains: False
Threshold: 6
Minimum Area: 15
Block Size: 179

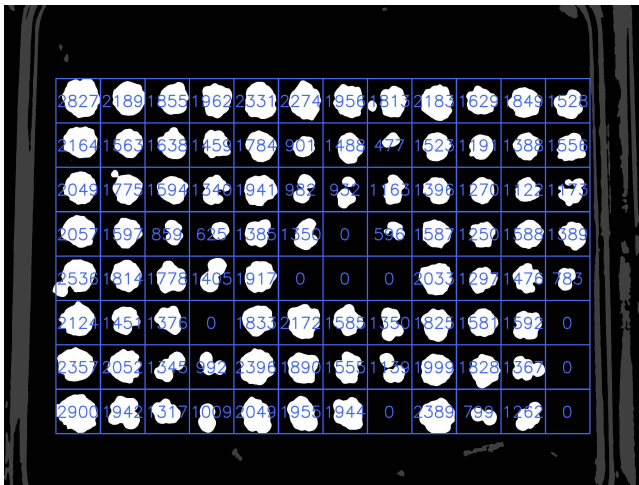
Preview Image



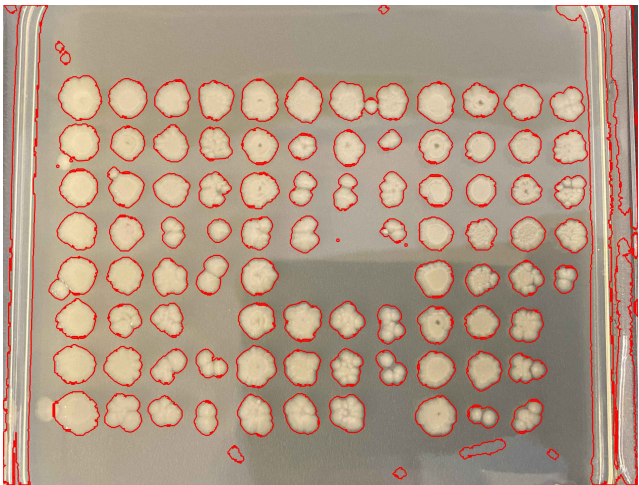
Tool Usage: Red(+) Blue(-)



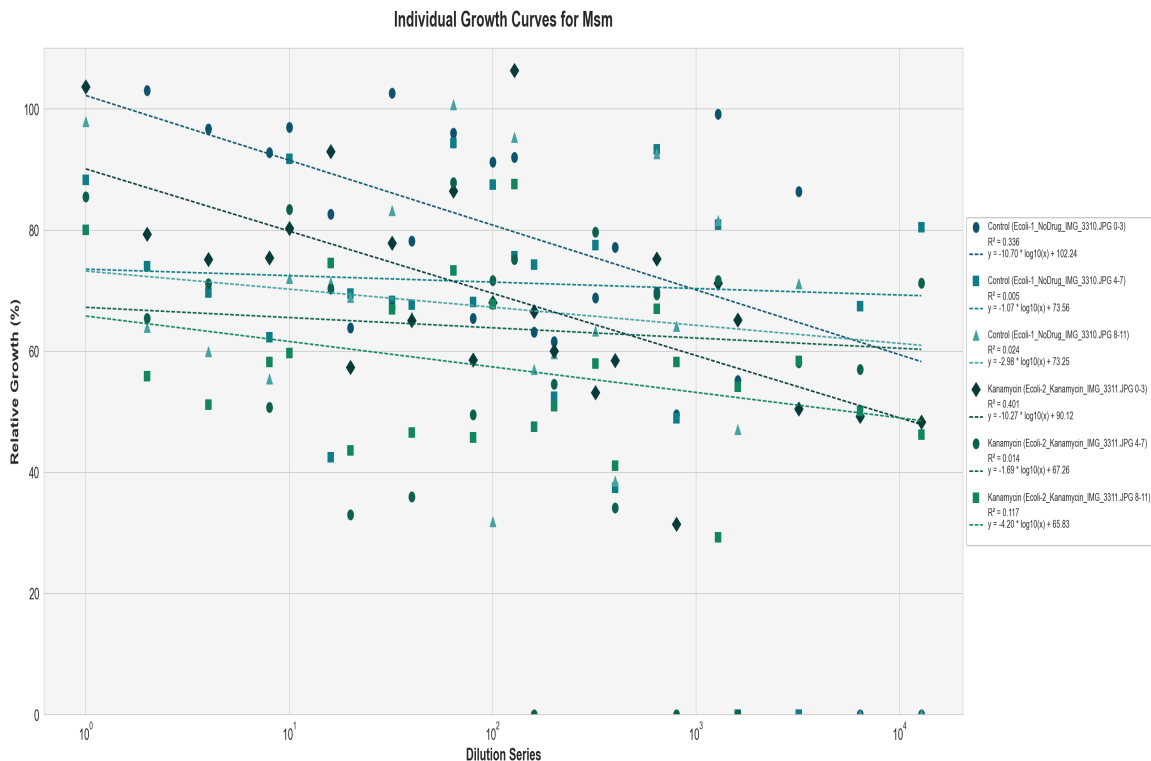
Binary Image



Grid Image

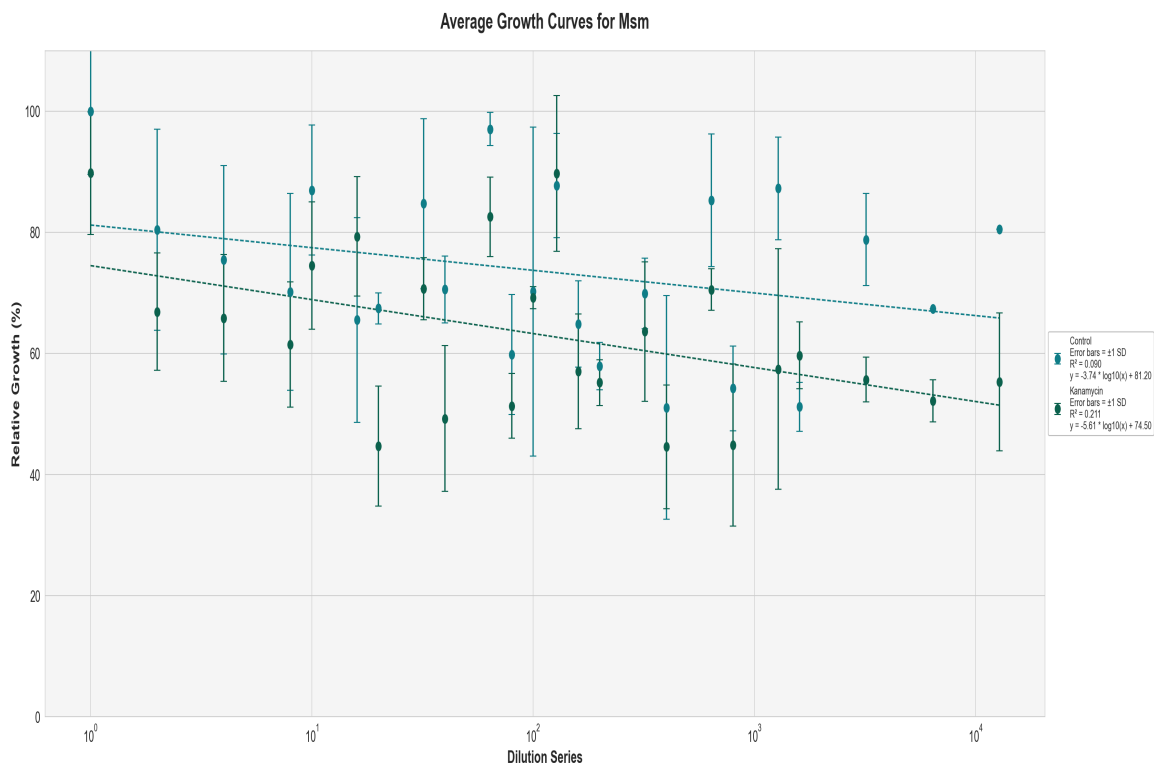


Contours Image



Control (Ecoli-1_NoDrug_IMG_3310.JPG 0-3)	Control (Ecoli-1_NoDrug_IMG_3310.JPG 4-7)	Control (Ecoli-1_NoDrug_IMG_3310.JPG 8-11)	Kanamycin (Ecoli-2_Kanamycin_IMG_3311.JPG 0-3)
Formula	$y = -10.70 * \log_{10}(x) + 102.24$	$y = -1.07 * \log_{10}(x) + 73.56$	$y = -2.98 * \log_{10}(x) + 73.25$
Slope	-10.70	-1.07	-2.98
Intercept	102.24	73.56	73.25
R-squared	0.336	0.005	0.024
y-cut	102.24	73.56	73.25
x-cut	3.61e+09	1.02e+69	3.53e+24
x_at_y50	76461.74	1.27e+22	6.19e+07

Kanamycin (Ecoli-2_Kanamycin_IMG_3311.JPG 4-7)	Kanamycin (Ecoli-2_Kanamycin_IMG_3311.JPG 8-11)
Formula	$y = -4.20 * \log_{10}(x) + 65.83$
Slope	-4.20
Intercept	65.83
R-squared	0.117
y-cut	65.83
x-cut	4.61e+15
x_at_y50	5850.60



	Control	Kanamycin
Formula	$y = -3.74 * \log_{10}(x) + 81.20$	$y = -5.61 * \log_{10}(x) + 74.50$
Slope	-3.74	-5.61
Intercept	81.20	74.50
R-squared	0.090	0.211
y-cut	81.20	74.50
x-cut	5.23e+21	1.90e+13
x_at_y50	2.21e+08	23278.58

This report was generated by **Spotplotter** version **1.0** on **2024-11-21**

Spotplotter was created by **Holly Lewis** with supervision from **R Verrinder** and **Dr. M Mason** as BSc (Eng) final year project submitted in partial fulfilment of the requirements for the degree of Bachelor of Science in Electrical and

To read the full report please see: [GITHUB LINK](#)

Formula: The formula represents the linear regression equation that models determined using the `linregress` function from `scipy.stats` which determines the relationship between the logarithm of the dilution series and relative growth. It follows the form: $y = m \cdot \log_{10}(x) + b$

Normalization: The relative growth values were normalized to a baseline to make the results comparable across different conditions. The quantified values of each spot were divided by the average value of the first spot in the -ATP for each strain.

The slope (m): indicates the rate of change in relative growth as the dilution series increases (on a logarithmic scale). A steep slope indicated that the growth has a faster knockdown as the dilution changes. A shallow slope indicates that the growth is more stable across dilutions.

The intercept (b) and y-cut is the relative growth when the solution is not diluted.

The R-squared value measures how well the linear regression line fits the data, where 1 represents a perfect fit and 0 represents no relationship. Higher R^2 values indicates that the growth follows a linear relationship.

The X-cut refers to the point where the regression line crosses the x-axis, indicating the dilution value at which the relative growth would theoretically be zero (no growth).

X at Y = 50: This value represents the dilution series value when the relative growth is 50% i.e., the knockdown is 50% in comparison to the -ATP series.